

COMP 3270 – Introduction to Algorithms

Homework 3

Topics – time complexity

1. Prove that the running time of an algorithm is $\theta(g(n))$ if and only if its worst-case running time is $O(g(n))$ and best-case running time is $\Omega(g(n))$.
2. Which is asymptotically larger: $\lg n$ or $\sqrt{n}$? Explain your answer.
3. Prove that $e^{1/n} \in O(n^t), t > 0$
4. For each of the following functions, which of the following asymptotic bounds hold for $f(n)$: $O(g(n)), \Omega(g(n)),$ or $\Theta(g(n))$?
 - a. $f(n) = 3n^2, g(n) = n^2$
 - b. $f(n) = 2n^4 - 3n^2 + 7, g(n) = n^5$
 - c. $f(n) = \log n, g(n) = \log n + 1/n$
 - d. $f(n) = 2^{k \log n}, g(n) = n^k$
 - e. $f(n) = 2^n, g(n) = 2^{2n}$
5. Use formal definitions to prove that $\sqrt{(n+1)^3} = \Omega(n\sqrt{n})$, hint $\sqrt{ab} = \sqrt{a}\sqrt{b}$ for any non-negative real numbers a and b .